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Article in *Journal of Applied Microbiology* · August 2020

DOI: 10.1111/jam.14822

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ORIGINAL ARTICLE

Quinolone resistance among *Salmonella* Kentucky and Typhimurium isolates in Tunisia: first report of *Salmonella* Typhimurium ST34 in Africa and *qnrB19* in Tunisia

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Keywords

DNA sequencing, pulsed-field gel electrophoresis, quinolone resistance, *Salmonella*, Tunisia.

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2020/1124: received 2 June 2020, revised 28 July 2020 and accepted 7 August 2020

doi:10.1111/jam.14822

Abstract

Aims: Characterization of quinolone-resistant *Salmonella* Kentucky and Typhimurium isolates in Tunisia from various sources, detection of some plasmid-mediated quinolone resistance genes and the genetic relatedness.

Methods: A total of 1404 isolates of *S. Kentucky* ($n = 1059$)/*S. Typhimurium* ($n = 345$) from various sources from all over Tunisia were tested for quinolone resistance by disk diffusion method. Minimum inhibitory concentrations of nalidixic acid, ciprofloxacin and ofloxacin were determined. Quinolone-resistant isolates were screened for plasmid-mediated quinolone-resistance genes (*qnrA*, *qnrB*, *qnrS*, *aac(6')-Ib-cr* and *qepA*) by polymerase chain reaction (PCR). Mutations in the quinolone-resistance-determining regions of the *gyrA* and *parC* genes were detected by PCR and DNA sequencing. Pulsed-field gel electrophoresis and multilocus sequence typing were accomplished for isolates harbouring plasmid-mediated quinolone-resistance genes.

Results: According to our selection criteria (NAL = resistance phenotype; CIP = resistant with diameter 0, or intermediate), only 63 *S. Kentucky*/41 *S. Typhimurium* isolates were investigated: 49% (5/104) were multidrug resistant. Two *S. Typhimurium* isolates harboured *qnrB19* with different PFGE profiles. A mutation was detected in the *gyrA* gene for each of these two isolates. MLST revealed the presence of ST313 and ST34, an endemic sequence type.

Conclusion: Our study highlights the presence of quinolone multidrug-resistant *Salmonella* in humans and animals in Tunisia. This is the first report of *S. Typhimurium* ST34 in Africa and *qnrB19* in Tunisia.

Significance and Impact of the Study: This is the first report that describes not only the current epidemiological situation of the quinolone resistance in *S.*

Kentucky and Typhimurium isolated from various sources and regions in Tunisia, but also, the genetic resistance determinants associated with phenotypic antibiotic resistance and the molecular mechanisms of their quinolone-resistance. Also, we provide the first report of *S. Typhimurium* ST34 in Africa, and the first report of *qnrB19* in *Salmonella* in Tunisia.

Introduction

Salmonella is considered as a global public health concern and causes 93.8 million cases of gastroenteritis each year (Majowicz *et al.* 2010). Eighty millions of these cases were foodborne, resulting in the death of nearly 155 000 persons per year (Majowicz *et al.* 2010; Turki *et al.* 2012; Frasson *et al.* 2016). This situation is furthermore aggravated by the emergence of strains that are resistant to antibiotics. Multidrug-resistant *Salmonella* has been recognized as a major public health problem since the beginning of the 1990s (Browne *et al.* 2020). The increase in antimicrobial resistance in *Salmonella* is mainly caused by excessive use of antibiotics in human and veterinary medicine (Wang *et al.* 2019). In response to the excessive or even the misuse of antibiotics, *Salmonella* has developed many mechanisms that allow simultaneous resistance to various classes of antibiotics, leading to the loss of effective treatment options (Laws *et al.* 2019). Limited information is available on the occurrence of the emerging *Salmonella* antibiotic resistance in Tunisia (Guedda *et al.* 2009; Abbassi-Ghozzi *et al.* 2012; Turki *et al.* 2012; Al-Gallas *et al.* 2013; Oueslati *et al.* 2016; Ben Salem *et al.* 2017).

Fluoroquinolones are one of the last line treatments for multidrug-resistant *Salmonella* infections in adults (Browne *et al.* 2020). Unfortunately, fluoroquinolone-resistant zoonotic and environmental *Salmonella* isolates have frequently been reported and have become increasingly common in Europe and the United States (Turki *et al.* 2012; Le Hello *et al.* 2013; Wasyl *et al.* 2015; Acheampong *et al.* 2019). In recent years, the Kentucky serotype emerged in Egypt and spread throughout Africa and the Middle East (between 2002 and 2008) (Le Hello *et al.* 2013).

Quinolone resistance in Enterobacteriaceae is mostly mediated by point mutations in the quinolone-resistance-determining regions (QRDR) of the DNA gyrase (*gyrA* and *gyrB*) and topoisomerase IV genes (*parE* and *parC*), leading to target modification. In *Salmonella* spp., these mutations are related to resistance to nalidixic acid (NAL) and reduced susceptibility to fluoroquinolones such as that of ciprofloxacin (CIP) (Cavaco and Aarestrup 2009). Also, plasmid-mediated quinolone resistance (PMQR) has emerged in *Salmonella* sp. and other Enterobacteriaceae with increasing prevalence (Pham *et al.* 2019; Vieira *et al.* 2019).

Currently, three mechanisms are recognized as PMQRs: The *qnr* genes with five different *qnr* families, each with different numbers of alleles (*qnrA1*–7, *qnrS1*–4, *qnrB1*–31, *qnrC*, and *qnrD*) (Jacoby *et al.* 2009), a modified aminoglycoside acetyl-transferase gene (*aac(6′)-Ib-cr*) (Robicsek *et al.* 2005), and a specific quinolone efflux pump (*qepA*) (Yamane *et al.* 2007).

Ciprofloxacin is the drug of choice in developing countries due to its low cost, oral route of administration and easy availability (Sharma *et al.* 2017). However, strains of *Salmonella* with increased levels of resistance to fluoroquinolones and treatment failures due to decreased susceptibility to ciprofloxacin have been reported (Howard *et al.* 1990; Pers *et al.* 1996; Vasallo *et al.* 1998; Aarestrup *et al.* 2003; Acheampong *et al.* 2019). Importantly, isolates with ciprofloxacin minimum inhibitory concentrations (MICs) as low as 0.125 µg ml^{−1} have been associated with treatment failure (Aarestrup *et al.* 2003; Gai *et al.* 2019).

Based on susceptibility testing results of NAL and CIP, quinolone resistance can be ranked at three levels: resistant to CIP (resistant to both drugs), or reduced CIP susceptibility (resistant to nalidixic acid and intermediate to CIP), and susceptible to both drugs (Hassing *et al.* 2013). Recently, concern has been raised about highly CIP-resistant strains of several *Salmonella* serotypes circulating in both human beings and the environment (Gong *et al.* 2016). To treat salmonellosis cases caused by CIP-resistant *Salmonella* strains, a third-generation cephalosporin is the first choice (Dutta *et al.* 2001; Guiral *et al.* 2019).

According to the National Reference Center for *Salmonella*, *Shigella*, and *Vibrio* sp. at the Pasteur Institute of Tunis, *S. Kentucky* and *S. Typhimurium* are ranked as the second and third most important serovars in Tunisia, respectively. Thus, this study will focus only on these two serotypes. All the isolates included in this study should be screened for nalidixic acid resistance to avoid false reporting of ciprofloxacin susceptibility. Furthermore, the determination of the MIC of ciprofloxacin is essential to assess the true susceptibility pattern.

The aim of this study was the characterization of quinolone-resistant *Salmonella* Kentucky and Typhimurium isolates in Tunisia (2010–2014) from various sources. Also, we examined the occurrence of some PMQR genes, and establish the genetic relatedness of these isolates.

Materials and methods

Salmonella isolates

This is a retrospective study conducted on a total of 1404 isolates conserved in the National Center of *Salmonella*, *Shigella*, and *Vibrio* spp-Enteropathogens-Institut Pasteur de Tunis, during the years 2010–2014: *S. enterica* serovar Kentucky ($n = 1059$) and *S. Typhimurium* ($n = 345$) recovered from food (Kentucky = 300, Typhimurium = 50), human (Kentucky = 260, Typhimurium = 120), animal (Kentucky = 400, Typhimurium = 100) and environmental (Kentucky = 99, Typhimurium = 75) samples. The isolates were grown for 18–24 h in Luria-Bertani broth at 37°C. A standard technique was used to isolate *Salmonella* strains in all laboratories all over Tunisia throughout the study period (Ben Aissa *et al.* 2007). Then, all isolates were stored in –80°C until analysis.

Confirmation and serotyping were done in the National Center of *Salmonella*, *Shigella*, *Vibrio*-Enteropathogens—Institut Pasteur de Tunis, according to the Kauffman–White scheme. Serotyping was performed by slide agglutination test to identify the somatic O antigens and flagellar H antigens (Grimont and Weill 2007) (Bio-Rad, Marnes-la-Coquette, France).

Antimicrobial susceptibility testing

Disk diffusion with Mueller–Hinton agar was performed for the antimicrobial susceptibility test. The interpretations of the results were carried out according to the guidelines of the Antibiogram Committee of the French Society for Microbiology 2015 (French Society of Microbiology 2015) (http://www.sfm-microbiologie.org/UserFile/s/files/casfm/CASFM_EUCAST_V1_2015.pdf) (CA-SFM 2015). *Escherichia coli* ATCC 25922 was used as a control.

A total of 23 antibiotics were tested: amoxicillin (AMX, 25 µg), amoxicillin + clavulanic acid (AMC, 20/10 µg), ampicillin (AMP, 10 µg), ticarcillin (TIC, 75 µg), ceftazidime (CAZ, 10 µg), cefotaxime (CTX, 5 µg), cefoxitin (FOX, 30 µg), cephalothin (CEF, 30 µg), ceftriaxone (CRO, 30 µg), aztreonam (ATM, 30 µg), imipenem (IPM, 10 µg), nalidixic acid (NAL, 30 µg), ofloxacin (OFX, 5 µg), ciprofloxacin (CIP, 5 µg), kanamycin (KMN, 30 µg), gentamicin (GEN, 10 µg), amikacin (AMK, 30 µg), streptomycin (STR, 10 UI), sulphonamide (SSS, 200 µg), trimethoprim–sulfamethoxazole (SXT, 1-25/23-75 µg), chloramphenicol (CHL, 30 µg), tetracycline (TET, 30 µg) and azithromycin (AZM, 15 µg).

A quinolone-resistant phenotype was defined as resistant to at least one antibiotic from the quinolone family.

In our study, we used three antibiotics from this family: NAL, CIP and OFX.

The MICs of nalidixic acid, ciprofloxacin and ofloxacin were determined by the agar dilution method. The MIC breakpoints used for susceptibility and resistance to nalidixic acid were ≤ 8 and ≥ 16 µg l⁻¹, respectively, as recommended by the Antibiogram Committee of the French Society for Microbiology (French Society of Microbiology 2015) (CA-SFM 2015). The breakpoints for interpreting NAL disk diffusion zone diameters were ≥ 19 mm (S), 14–18 mm (I) and ≤ 13 mm (R).

For ciprofloxacin and ofloxacin isolates, MICs were classified as recommended by the European Committee on Antimicrobial Susceptibility Testing (The European Committee on Antimicrobial Susceptibility Testing 2014) (http://www.eucast.org/clinical_breakpoints/). For CIP, ≤ 0.064 µg l⁻¹ as susceptible (S), 0.12–0.5 µg l⁻¹ as intermediate (I) and ≥ 1.0 µg ml⁻¹ as resistant (R). For OFX, ≤ 0.125 µg l⁻¹ as (S), 0.25–1 µg ml⁻¹ as (I) and ≥ 2 µg ml⁻¹ as (R). The breakpoints for interpreting disk diffusion zone diameters were ≥ 31 mm (S), 21–30 mm (I) and ≤ 20 mm (R) for CIP; and ≥ 25 mm (S), 16–24 mm (I) and ≤ 15 mm (R) for OFX.

Screening for plasmid-mediated quinolone-resistance genes

Only 104 isolates ($n = 63$ *S. Kentucky*, $n = 41$ *S. Typhimurium*) were studied to detect plasmid-mediated quinolone-resistance (PMQR) genes. According to the selection criteria, NAL = resistance phenotype; CIP = resistant with diameter 0, or intermediate, OFX = resistance phenotype.

For rapid DNA preparation, a single colony was transferred to distilled water and boiled at 100°C for 5 min followed by a centrifugation step, as previously described (Al-Gallas *et al.* 2002).

The topoisomerase genes (*parC* and *gyrA* genes) were amplified for isolates that harboured PMQR genes. The amplification was done by simplex polymerase chain reaction (PCR), as previously described (Griggs *et al.* 1996; Eaves *et al.* 2004). The *qnrA*, *qnrB* and *qnrS* genes encoding for PMQR were amplified by a multiplex PCR, as described previously (Wang *et al.* 2008). The presence of *qepA* and *aac(6′)-Ib-cr* genes were analysed by simplex PCR (Park *et al.* 2006; Ruiz *et al.* 2012). The primers used for PCR and DNA sequencing are listed in Table 1. Positive controls for PCR experiments were provided by Dr Abbassi from the Veterinary Research Institute of Tunisia. *E. coli* strains were used as control for detection of *qnr* and *acc(6′)-Ib-cr* genes, whereas *S. Typhimurium* for *qepA* gene (Al-Gallas *et al.* 2013).

Table 1 List of primers used in this study

Gene	Primer (5'–3')	T _m	Amplicon	Reference
<i>parC</i>	CTATGCGATGTCAGAGCTGG TAACAGCAGCTCGGCGTATT	55°C	270 bp	Eaves et al. (2004)
<i>gyrA</i>	TGTCCGAGATGGCCTGAAGC TACCGTCATAGTTATCCACG	64°C	347 bp	Griggs et al. (1996)
<i>qnrB</i>	GATCGTGAAAGCCAGAAAGG ACGATGCCTGGTAGTTGTCC	53°C	496 bp	Wang et al. (2008)
<i>qnrA</i>	ATTCTCACGCCAGGATTG GATCGGCAAAGGTTAGGTCA	53°C	516 bp	Wang et al. (2008)
<i>qnrS</i>	ACGACATTCGTCAACTGCAA TAAATTGGCACCTGTAGGC	53°C	417 bp	Wang et al. (2008)
<i>qep A</i>	GGACATCTACGCTTCTTCG CAACTGCTTGAGCCCGTAG	60°C	617 bp	Rocha-Gracia et al. (2010)
<i>aac(6')-Ib</i> variants	TTGCGATGCTCTATGAGTGGCTA CTCGAATGCCTGGCGTGTTT	55°C	482 bp	Park et al. (2006) Then direct sequencing of the PCR product with primer 5'-CGTCACTCCATACATTGCAA-3'

Detection of mutations in QRDR and PMQR: DNA sequencing

Amplicons of *parC* and *gyrA* PCRs were sequenced for the strains that harboured PMQR genes. Sequencing was done using a BigDye Terminator ver. 3.1 Cycle Sequencing Kit (Applied Biosystems, Waltham, MA) and an ABI Prism 3100 sequencer (Applied Biosystems, Warrington, Cheshire, UK). Data collections were performed on the ABI PRISM 3100 Genetic Analyzer Data Collection Software ver. 1.1 (Applied Biosystems). We used BlastX to search for nucleotide databases using a nucleotide query. For *qnrB*, we used an e-value cut-off of 0.0 with an identity percentage = 100% and coverage = 100% (the aligned length is 100% of the total length of the domain). For *parC* and *gyrA*, we used an e-value cut-off of 3e-108 with an identity percentage of 100% and coverage of 100%. Sequences were compared with reference sequences from NCBI GenBank and deposited in GenBank under the accession numbers given in Table 5.

Molecular typing by pulsed-field gel electrophoresis

Pulsed-field gel electrophoresis (PFGE) was performed to generate genomic DNA fingerprinting profiles for *Salmonella* strains harbouring PMQR genes. To determine clonal relatedness among *Salmonella* strains, we analysed their *Xba I* patterns by PFGE according to Pulse Net (<https://www.cdc.gov/pulsenet/pdf/ecoli-shigella-salmonella-pfge-protocol-508c.pdf>) and as described elsewhere (Al-Gallas et al. 2013).

The relatedness of different patterns was analysed by cluster analysis using the Dice coefficient for band

matching with 1% position tolerance and the unweighted pair group method with arithmetic average.

Multilocus sequence typing

Multilocus sequence typing (MLST) was carried out using the MLST protocols provided on the MLST website (http://mlst.ucc.ie/mlst/dbs/Senterica/documents/primerEnterica_html). A minimum spanning tree was created using Bionumerics software 6.5 according to the instructions (the unweighted pair group method of arithmetic averages method). Primer pairs (Table 2; IDT, Coralville, IA) were used to amplify the DNA for the presence of *thrA*, *purE*, *aroC*, *dnaN*, *hemD*, *hisD* and *sucA*. Sequence data obtained were imported into DNASTar (Lasergene, Madison, WI), trimmed, aligned to the control sequences obtained from the MLST site and interrogated against the MLST database. The generated sequence types were recorded and added to the strain information.

Supporting information

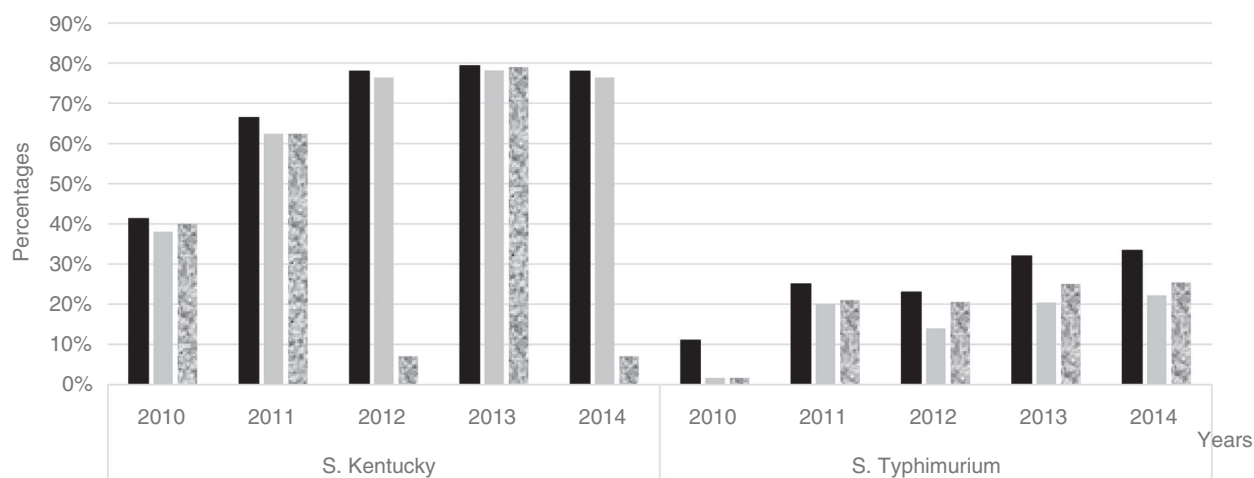
Antibiotic-resistance profiles for *Salmonella* Kentucky and *Salmonella* Typhimurium isolates of this study (2010–2014) are given in Table S1.

Results

Among 1404 tested *Salmonella* isolates (Kentucky = 1059; Typhimurium = 345), 59.7% (744 Kentucky and 94 Typhimurium) of the isolates were quinolone resistant (Fig. 1). There was an increase in the quinolone resistance during the 5 years for both studied serotypes.

Table 2 MLST-PCR primers and sequence primers used in the amplification and sequencing of the 'house-keeping' genes and product sizes

Gene	MLST-PCR primers	Amplicon bp	MLST-Seq primers	Amplicon bp
<i>thrA</i>	GTCACGGTGATCGATCCGGT	852	ATCCCGGCCGATCACATGAT	501
<i>thrA</i>	CACGATATTGATATTAGCCCG		CTCCAGCAGGGGCTCTTTCAG	
<i>purE</i>	ATGTCTTCCCGCAATAATCC	510	CGCATTATCCGGCGCGTGT	399
<i>purE</i>	TCATAGCGTCCCCGCGGATC		CGCGGATCGGGATTTTCCAG	
<i>aroC</i>	CCTGGCACCTCGCGCTATAC	826	GGCACCAGTATTGGCCTGCT	501
<i>aroC</i>	CCACACACGGATCGTGGCG		CATATGCGCCACAATGTGTTG	
<i>dnaN</i>	ATGAAATTTACCGTTGAACGTGA	833	CCGATTCTCGGTAACCTGCT	501
<i>dnaN</i>	AATTTCTCATTGAGAGGATTGC		CCATCCACCAGCTTCGAGGT	
<i>hemD</i>	GAAGCGTTAGTGAGCCGTCTGCG	666	GTGGCCTGGAGTTTCCACT	432
<i>hemD</i>	ATCAGCGACCTTAATATCTTGCCA		GACCAATAGCCGACAGCGTAG	
<i>hisD</i>	GAAACGTTCCATTCCGCGCAGAC	894	GTCGGTCTGTATATCCCGG	501
<i>hisD</i>	CTGAACGGTCATCCGTTTCTG		GGTAATCGCATCCACCAAATC	
<i>sucA</i>	AGCACCGAAGAGAAACGCTG	643	AGCACCGAAGAGAAACGCTG	643
<i>sucA</i>	GGTTGTTGATAACGATACGTAC		GGTTGTTGATAACGATACGTAC	

**Figure 1** Changing in the quinolone resistance (■ NAL; ■ CIP; ■ OFX).

However, only for *S. Kentucky*, a variation in resistance for OFX was observed.

According to our selection criteria (NAL = resistance phenotype; CIP = resistant with diameter 0, or intermediate), only 104 isolates (*S. Kentucky* = 63 isolates (animal = 19, environment = 17, food = 16, human = 11), *S. Typhimurium* = 41 isolates (human = 19, food = 12, environment = 5, animal = 5)) were selected for the investigation. The antibiotic susceptibility of these isolates (Table 3) showed high resistance to OFX, CEF and TIC for *S. Kentucky*, and TIC, AMX and AMP for *S. Typhimurium*.

In *S. Kentucky* isolates, resistance to the quinolones were found (NAL 100% (63/63), OFX 75.0% (47/63) and CIP 63.5% (40/63)) (Table 3). Human isolates showed the highest rates of resistance against OFX and CIP, with 83.3% (10/12) followed by food isolates (OFX 81.3% (13/

16) and CIP (68.8%, 11/16)). In isolates from animals, CIP (57.9%, 11/19) and OFX resistance were detected (63.2%, 12/19). Environmental isolates showed 68.2% (11/16) and 56.3% (9/16) resistance for OFX and CIP, respectively (Fig. 2a).

In *S. Kentucky* isolates, resistance to β -lactam antibiotics and other antibiotics in association with quinolone resistance was detected (Table S1).

Selected *S. Typhimurium* isolates were resistant not only against quinolones (NAL 100% (41/41); OFX 39% (16/41); CIP 19.5% (15/41)) but also against the β -lactam family (AMX, AMP, TIC with 56.1% (23/41) for each) (Table 3). Resistance to other antibiotics was also shown (Table 3). No resistance was detected for CZD, FOX, CRO, ATM, GMN or AMK.

Salmonella Typhimurium isolates from food-animal products (29.3% (12/41)) were mainly recovered from

Table 3 Antibiotic susceptibility of *Salmonella* strains included in this study (104 strains)

Antibiotic	Total no. of <i>S. Kentucky</i> isolates = 63		Total no. of <i>S. Typhimurium</i> isolates = 41	
	Resistance no. (%)	Sensitive no. (%)	Resistance no. (%)	Sensitive no. (%)
AMP	32 (51)	31 (49.2)	21 (34)	20 (49)
AMX	27 (43)	36 (57)	23 (37)	18 (44)
AMC	2 (3.2)	61 (97)	3 (2.4)	38 (93)
TIC	43 (68.2)	20 (32)	23 (37)	18 (44)
CEF	45 (71.4)	18 (28.6)	8 (17)	33 (80.5)
FOX	1 (1.6)	62 (98.4)	0	41 (100)
COX	1 (1.6)	62 (98.4)	1 (2.4)	40 (97.6)
CZD	0	63 (100)	0	41 (100)
CRO	0	63 (100)	1 (2.4)	40 (97.6)
IPM	0	63 (100)	0	41 (100)
ATM	1 (1.6)	62 (98.4)	0	41 (100)
SMN	8 (13)	55 (87.3)	13 (19.5)	28 (68.3)
KMN	10 (16)	53 (84)	5 (10)	36 (88)
AMK	0	63 (100)	0	41 (100)
GMN	4 (6.3)	59 (94)	0	41 (100)
OFX	47 (75)	16 (25.4)	16 (39)	25 (61)
CIP	40 (63.5)	23 (36.5)	15 (19.5)	26 (63.4)
NAL	63 (100)	0	41 (100%)	0
SSS	7 (11)	56 (89)	17	24 (58.5)
SXT	3 (5)	60 (95.2)	3 (5)	38 (93)
CHL	0	63 (100)	0	41 (100)
TET	0	63 (100)	0	41 (100)
AZT	6 (9.5)	57 (90.5)	3 (7.3)	38 (93)

AMX: amoxicillin; AMC: amoxicillin + clavulanic acid; AMP: ampicillin; TIC: ticarcillin; CZD: ceftazidime; COX: cefotaxime; FOX: cefoxitin; CEF: cephalothin; CRO: ceftriaxone; ATM: azetreonam; IPM: imipenem; NAL: nalidixic acid; OFX: ofloxacin; CIP: ciprofloxacin; KMN: kanamycin; GMN: gentamicin; AMK: amikacin; SMN: streptomycin; AZT: azithromycin; SXT: trimethoprim-sulfamethoxazole; CHL: chloramphenicol; TET: tetracycline; SSS: sulphonamide.

chicken (12.2% (5/41)), red meat (7.3% (3/41)) and other sources (10% (4/41)). The highest levels of resistance against OFX and CIP were found in food (66.7%, 8/12) and animal isolates (50.0%, 3/6) (Fig. 2b). The second highest rates of resistance against OFX (50.0%, 3/6) and CIP (33.3%, 4/12) were detected in animal and food isolates, respectively. Human isolates were resistant to OFX and CIP with 27.8% (5/18) for each (Fig. 2b).

Multidrug resistance (MDR) was defined as acquired non-susceptibility to at least one agent in three or more antimicrobial categories (Basak et al. 2016). Overall, 49% (51/104) of *Salmonella* isolates showed MDR; *S. Kentucky* and *S. Typhimurium* 55% (28/51) and 45% (23/51), respectively. In all, 33 different MDR profiles from 51 different resistance profiles were observed (Table S1). In all, 51 (49%) isolates were multidrug resistant: 31.4% (16/51) of human isolates, 23.5% (12/51) of food isolates, 22% (11/51) of environment isolates and 20% (10/51) of

animal isolates. Animal and food MDR isolates were mainly recovered from poultry. Most MDR profiles in our collection were from stool samples.

The MICs of NAL, CIP and OFX for *Salmonella* isolates are given in Table 4. MICs of *Salmonella* isolates ranged from 32 to >2 mg l⁻¹ for NAL, CIP and OFX respectively.

Concerning PMQR genes and after screening all 104 strains by PCR, only two *S. Typhimurium* isolates (no. 713/2014 region Monastir, no. 92/2013 region Tunis) were recovered from stool samples of human patients harbouring the *qnrB* gene (Fig. 3). Both isolates did not show resistance to CIP. However, both showed MDR (AMX, AMP, TIC, NAL, GMN, SMN, TET and SSS). The *aac(6')-Ib-cr* gene was observed in one *S. Typhimurium* (no. 92/2013 region Tunis) (Fig. 4). Only these two isolates were sequenced for detecting mutations in the *gyrA* and *parC* genes. The sequencing of the *gyrA* gene of these two isolates revealed chromosomal substitutions at two positions (Ser 83 >Glu; Asp 87 >Arg). However, the sequencing of the *parC* gene of these two isolates did not reveal mutations.

When we blast the sequence of *qnrB* of these two strains (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>), we obtained a high alignment score. We obtained a query coverage of 100%, with few sequences producing significant alignments. Both isolates were identified as *qnrB19* (Table 5).

The two *S. Typhimurium* isolates were investigated by PFGE but were not closely related. Both belonged to *S. Typhimurium* ST34 (strain code: 713/2014) and *S. Typhimurium* ST 313 (strain code: 1215/2013) (Table 5).

Discussion

To our knowledge, no Tunisian study has focused on quinolone resistance and genotyping in *Salmonella Kentucky*.

In this work, we evaluated the quinolone resistance in *Salmonella enterica* serovar Kentucky and Typhimurium recovered from various sources (food, humans, animals and the environment) during the years 2010–2014. The resistance rate to quinolones in this study was higher than that reported in our latest study in 2013, where only 2.8% (2/70) of *S. Typhimurium* were quinolone resistant (Al-Gallas et al. 2013). We examined the DNA sequencing of *gyrA* and *parC* to detect amino acid substitution of the quinolone resistance in selected isolates and determined their genetic relatedness.

The spread of quinolone-resistant *Salmonella* isolates in humans is a major health risk, especially in immunocompromised humans that might need antimicrobial therapy (Kuang et al. 2015).

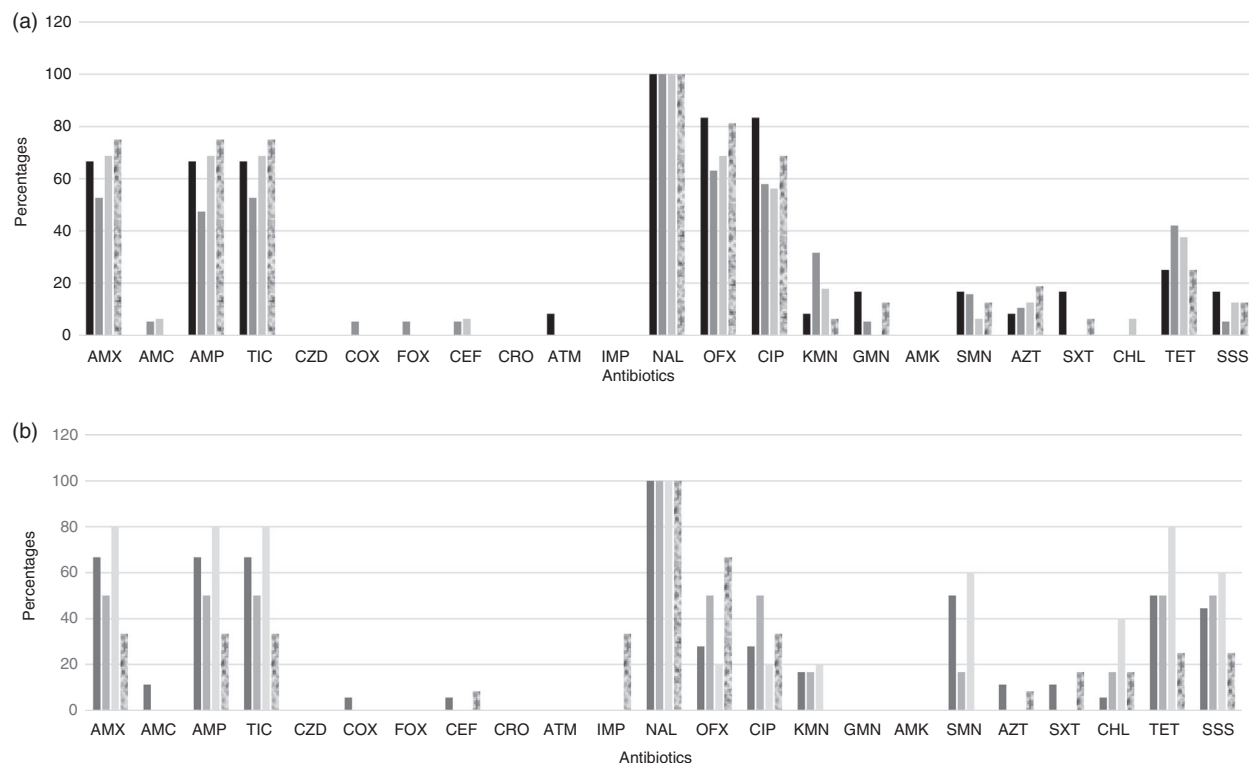


Figure 2 Antibiotic resistance for quinolone resistance *Salmonella* Kentucky (a) and *Salmonella* Typhimurium (b) isolates investigated (2010–2014) (a: ■ human ($n = 12$); ■ animal ($n = 19$); ■ environment ($n = 16$); ■ food (16); b: ■ human ($n = 18$); ■ animal ($n = 6$); ■ environment ($n = 5$); ■ food (12)).

Table 4 MIC results for the 104 quinolone-resistant *Salmonella* isolates in this study

Serotype	MIC ($\mu\text{g ml}^{-1}$) NAL	No. of isolates	MIC ($\mu\text{g ml}^{-1}$) CIP	No. of isolates	MIC ($\mu\text{g ml}^{-1}$) OFX	No. of isolates	PMQR mechanism <i>qnrB</i> , <i>qnrA</i> , <i>qnrS</i>	QRDR <i>parC</i> , <i>gyrA</i>	<i>aac(6')-Ib-cr</i>
Kentucky	32	63	0.5	24	>2	63	–	<i>parC</i> , <i>gyrA</i>	–
Typhimurium	32	39	1	39	>2	17	–	<i>parC</i> , <i>gyrA</i>	–
			1	13			–	<i>parC</i> , <i>gyrA</i>	–
	16	1	1	1			<i>qnrB</i> ⁺	<i>parC</i> , <i>gyrA</i>	<i>aac(6')-Ib-cr</i>
	16	1	1	1			<i>qnrB</i> ⁺	<i>parC</i> , <i>gyrA</i>	–

MIC: minimum inhibitory concentration; NAL: nalidixic acid; CIP: ciprofloxacin; OFX: ofloxacin; PMQR: plasmid-mediated quinolone resistance; QRDR: quinolone-resistance-determining region.

The resistance rate to quinolones in this study was higher than that reported in our latest study where only 2.8% (2/70) of *S. Typhimurium* were quinolone resistant in 2013 (Al-Gallas *et al.* 2013). Concerning *S. Kentucky*, to our knowledge, there is no published report that has focused on quinolone resistance in this emerging serotype. Our findings confirm the increase of quinolone resistance in Tunisia over those years.

Theoretically, NAL resistance also confers cross-resistance to fluoroquinolone compounds due to an identical target site (Thu *et al.* 2019). All quinolone strains in this study were resistant to NAL.

The strains in this study showed a high rate of resistance to CIP. This is in contrast to various data from the Tunisian literature (Guedda *et al.* 2009; Turki *et al.* 2012; Al-Gallas *et al.* 2013), which reported resistance rates of <50%.

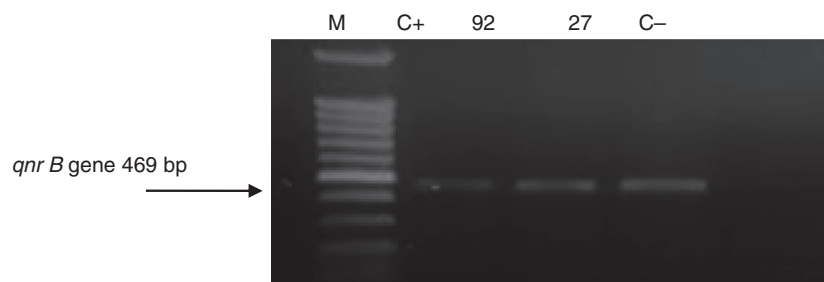


Figure 3 PCR amplification of qnrB products (469 pb) in *Salmonella* Typhimurium isolates.

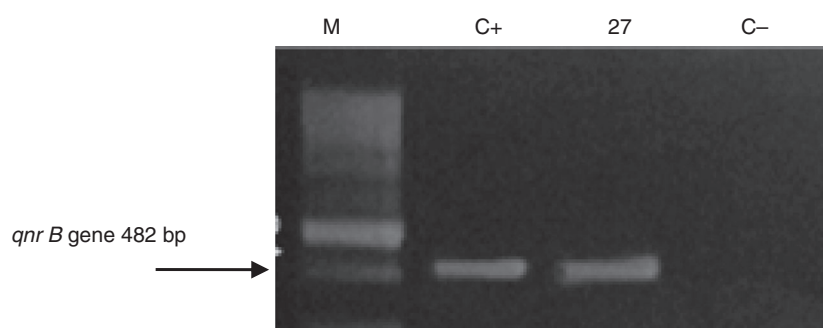


Figure 4 PCR amplification of aac(6')-Ib product (482 pb) in *Salmonella* Typhimurium isolates.

Table 5 Characteristics of the two *S. Typhimurium* isolates recovered from stool samples of human patients with PMQR mechanisms genes and topoisomerase genes for quinolone-resistance-determining region

PCR Code	Pasteur Code	Source	Date of isolation/Region	Resistant against antibiotic tested	Detected genes	Molecular typing MLST	Sequence accession number
27	713 /14	Stool	22/05/14 Monastir	AMX, AMP, TIC, COX, CEF, CRO, NAL, SMN, TE, SUL	<i>parC</i> +, <i>gyrA</i> +, <i>qnr B</i> +, <i>aac(6')-Ib-cr</i> +	ST34	<i>qnrB19</i> : MF615304 <i>parC</i> : MF627604 <i>gyrA</i> : MN193565
92	1215/13	Stool	15/07/13 Tunis	AMX, AMP, TIC, NAL, SMN, TE, SUL	<i>parC</i> +, <i>gyrA</i> +, <i>qnr B</i> +	ST313	<i>qnrB19</i> : MF615305 <i>parC</i> : MF627605 <i>gyrA</i> : MN193565

AMX: amoxicillin; AMP: ampicillin; TIC: ticarcillin; COX: cefotaxime; CEF: cephalothin; CRO: ceftriaxone; NAL: nalidixic acid; GMN: gentamicin; SMN: streptomycin; TET: tetracycline; SSS: sulphonamide.

Among the CIP-resistant *S. Kentucky*, human isolates showed the highest rate of resistance. These results are similar to those reported in a study in 2013, where 83% (376/455) of *S. Kentucky* were resistant to ciprofloxacin. However, in our previous work, we showed only 40% (200/497) resistance to ciprofloxacin between 2000 and 2008 (Al-Gallas *et al.* 2013). These variations in resistance to ciprofloxacin suggest that antimicrobial resistance is increasing.

Regarding *S. Typhimurium*, all quinolone-resistant isolates were originated from poultry products. High rates of resistance to penicillin antibiotics (AMX, AMP, and TIC) were observed in both serotypes, which can be

explained by the frequent use of these antibiotics, particularly AMP, in the treatment of salmonellosis since the 1980s (Daniel *et al.* 2016; Browne *et al.* 2020).

Salmonella Typhimurium showed high rates of resistance in this report. In fact, *S. Typhimurium* has long been known to be more resistant than other common serovars (Daniel *et al.* 2016; Browne *et al.* 2020).

Concerning MDR, the proportion of such isolates was high (nearly 49%) and greater than that reported in previous studies (Parveen *et al.* 2007; Mendonça *et al.* 2019).

In all, 28 out of 51 (55%) *S. Kentucky* isolates showed MDR. In fact, MDR and ciprofloxacin-resistant *S.*

Kentucky have been reported in African countries for a long time, but not in Tunisia. Importantly, the majority of foodborne *Salmonella* in this research was multidrug resistant and this finding is consistent with the results of a recent study in Morocco (Amajoud *et al.* 2017).

Based on our findings, the presence of a high percentage of MDR among quinolone-resistant *Salmonella* isolates in Tunisia can be explained by the irrational and excessive use of quinolones and other antibiotics in poultry production systems, as well as in irrigation systems. This leads to contamination of animal products and vegetables, having a severe impact on the health of the population; it can be speculated that this practice might even have adverse effects on the tourism industry (Jacoby *et al.* 2009; Oueslati *et al.* 2016; Ben Salem *et al.* 2017).

Quinolones target two enzymes linked to bacterial DNA replication: DNA gyrase and DNA topoisomerase IV. The subunits of DNA gyrase are encoded by *gyrA* and *gyrB*, and that of the topoisomerase IV is encoded by *parC* and *parE* (Bhatnagar and Wong 2019). Quinolone resistance is usually associated with alterations in *gyrA*, *parC* or both (Bhatnagar and Wong 2019). In fact, two chromosomal substitutions at position 83 changing serine to proline (TCA) → (CCA) and at position 87 (aspartic acid (GAC) to glycine (GGC)) were observed in the *gyrA* gene of two sequenced *S. Typhimurium* isolates. Our sequences for the amplified products of the *gyrA* and *parC* genes for the two strains that harboured PMQR genes were 100% identical to the sequences in GenBank (Dutta *et al.* 2001).

Although some substitutions in *parC* were described in other studies (Hsueh *et al.* 2004; Izumiya *et al.* 2005; Wang *et al.* 2017), our isolates did not show any mutations. Our sequences for the amplified product of the *parC* gene revealed complete identity with one sequence in GenBank of a *S. Typhimurium* isolate from poultry in 2016 in Egypt (accession number: KX859486).

Only two isolates harboured *qnrB* genes among the 104 isolates investigated. These isolates were both characterized as *qnrB19* sequence variants. These sequences showed complete identity only with six *S. Typhimurium* strains isolated in Poland from a turkey farm environment (Dutta *et al.* 2001). These results are similar to studies where the *qnrB19* variants were identified in *S. Typhimurium* recovered from human samples (Dionisi *et al.* 2009; Jacoby *et al.* 2014). This finding can be explained by the fact that genes were present in the environment, and *Salmonella* acquired them through plasmid exchange (Jacoby *et al.* 2014). This is the first report of *Salmonella qnrB19* detection in Tunisia (source: human faeces).

As we mentioned earlier, in the introduction section, NAL disc diffusion assay should be used as a tool to detect *Salmonella* isolates with decreased ciprofloxacin

susceptibility. This is according to the CLSI 2014 guidelines where isolates with MIC $\geq 1 \mu\text{g ml}^{-1}$ are resistant to CIP. So, these two quinolone-resistant *S. Typhimurium* isolates were sensible to CIP.

The two quinolone-resistant *S. Typhimurium* isolates had different MLST genotypes. Both strains were isolated from humans with diarrhoea. ST313 has been found to be predominant in Africa (Jain *et al.* 2018; Puyvelde *et al.* 2019). A strong association of this genotype with the human disease had been reported previously (Kingsley *et al.* 2009; Puyvelde *et al.* 2019). ST34 was first reported in 2005 in Europe and spread worldwide (Arai *et al.* 2018) but not in Africa.

To our knowledge, this is the first report that describes not only the current epidemiological situation of the quinolone resistance in *S. Kentucky* and *S. Typhimurium* isolated from various sources and regions in Tunisia but also the genetic resistance determinants associated with phenotypic antibiotic resistance and the molecular mechanisms of their quinolone resistance. We reported a high rate of quinolone resistance in *S. Kentucky* (88.8%). Also, we provide the first report of *S. Typhimurium* ST34 in Africa. Only two *S. Typhimurium* strains harboured *qnrB*. These genes were both identified as *qnrB19*, which is again the first report of *qnrB19* in *Salmonella* in Tunisia. The study highlights the prevalence of multidrug-resistant bacteria in humans and animals in Tunisia and underlines that the WHO and OIE action plans against antimicrobial resistance must be followed.

Acknowledgements

This work was supported by Programmes Collaboratifs Internes (PCI). It was funded by Institut Pasteur de Tunis (Grant number 2013-PCI09). Also, we thank American Manuscript Editors for language editing.

Conflict of Interest

No conflict of interest declared.

Author contributions the relevant CRediT roles

Nazek Al-Gallas: Methodology, supervision, investigation and writing the original draft; **Nadine Khadraoui:** investigation; **Helmut Hotzel:** MLST investigation, review and editing; **Herbert Tomaso:** MLST investigation; **Hosny El-Adawy:** validation; **Heinrich Neubauer:** visualization; **Khoulood Belghouthi:** investigation; **Kais Ghedira:** data curation; **Hemant K. Gautam:** review; **Bipul Kumar:** data curation; **Dhafer Laouini:** Project administration; **Sinda Zarrouk:** Project administration; **Mohamed Salah**

Abbassi: Methodology PCR; **Ridha Ben Aissa:** Head of department.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Antibiotic resistance profiles for *Salmonella* Kentucky and *Salmonella* Typhimurium isolates of this study (2010–2014).